

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO5– Naturalization (BL5,BL6)	Assessment:	Assessment Tool 1 – Constraint Based Problem Solving
Weightage:	30% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO5 Statement:	<i>Analyze computational problems using backtracking techniques and complexity classes such as P, NP, NP-Hard, and NP-Complete for combinatorial problem solving.</i>		
Student Name:	KS THARUNESHWAR	Reg. No.:	192572148
Section / Batch:	2025	Date:	29-08-26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Apply backtracking techniques systematically for combinatorial problem solving.
- Clearly classify problems under P, NP, NP-Hard, and NP-Complete categories wherever required.
- Provide proper justification, state-space representation, and complexity analysis for all solutions.
- Neat presentation and logical explanation are mandatory.

Question Type	Focus Area	Questions	Marks
Constraint Based Problem Solving	Analyze combinatorial problems using backtracking and complexity classes such as P, NP, NP-Hard, and NPComplete.	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONSTARINT BASED PROBLEM SOLVING



SIMATS Online Programming Lab
DAA PROGRAMMING



CO5_AT1_Constraint-Based Problem Solving
← Back

Language: Python C C++ Java

QUESTIONS

Q1
BackTracking and Tractability
50 marks

Q2
BackTracking and Tractability
50 marks

2/2 answered

Q1 BackTracking and Tractability
50 marks

Constraint-Based Puzzle Solver (General CSP / NP-Complete Problem)

Solve a generalized puzzle with multiple constraints such as placement rules and adjacency conditions. Clearly define all constraints and rules. Analyze how constraints reduce the search space. Develop a backtracking-based solution. Apply constraint propagation and pruning techniques. Justify correctness and discuss computational complexity.

Your Code

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <stdbool.h>
4
5 #define MAX_N 20
6
7 int board[MAX_N];
8 int solution_count = 0;
9 bool is_safe(int row, int col, int n) {
10     for (int i = 0; i < row; i++) {
11         if (board[i] == col || abs(board[i] - col) == abs(i - row))
12             return false;
13     }
14     return true;
15 }
16
17 void solve_nqueens(int row, int n) {
18     if (row == n) {
19         solution_count++;
20         return;
21     }
22     for (int col = 0; col < n; col++) {
23         if (is_safe(row, col, n)) {
24             board[row] = col;
25             solve_nqueens(row + 1, n);
26         }
27     }
28 }
29
30 int main() {
31     int n;
32     if (scanf("%d", &n) != 1 || n <= 0) {
33         n = 8;
34     }
35     solve_nqueens(0, n);
36     printf("Total Solutions for %d-Queens: %d\n", n, solution_count);
37     return 0;
38 }

```

Input

stdin input

Output

Total Solutions for 8-Queens: 92



SIMATS Online Programming Lab
DAA PROGRAMMING



CO5_AT1_Constraint-Based Problem Solving
← Back

Language: Python C C++ Java

QUESTIONS

Q1
BackTracking and Tractability
50 marks

Q2
BackTracking and Tractability
50 marks

2/2 answered

Q2 BackTracking and Tractability
50 marks

Multi-Constraint Knapsack Problem (NP-Hard Problem)

Select items to maximize value while satisfying multiple constraints such as weight and volume. Clearly define all constraints. Analyze how constraints interact and restrict feasible selections. Develop a backtracking-based solution. Apply pruning to eliminate infeasible combinations. Justify correctness and discuss NP-Hard complexity.

Your Code

```

1 #include <stdio.h>
2
3 #define MAX_ITEMS 50
4
5 int max_value = 0;
6
7 typedef struct {
8     int value;
9     int weight;
10    int volume;
11 } Item;
12
13 void knapsack(Item items[], int n, int index, int current_weight, int current_volume) {
14     if (current_weight > max_w || current_volume > max_v) {
15         return;
16     }
17     if (current_value > max_value) {
18         max_value = current_value;
19     }
20
21     if (index == n) {
22         return;
23     }
24     knapsack(items, n, index + 1,
25              current_weight + items[index].weight,
26              current_volume + items[index].volume,
27              current_value + items[index].value,
28              max_w, max_v);
29     knapsack(items, n, index + 1, current_weight, current_volume,
30              current_value, max_w, max_v);
31 }
32
33 int main() {
34     int n, max_w, max_v;
35     if (scanf("%d %d %d", &n, &max_w, &max_v) != 3) {
36         return 0;
37     }
38 }

```

Input

3 10 10
60 5 5
100 4 6
120 8 7

Output

Maximum Value: 120

Code of Conduct

I V Deepa Dharshini (192525213) that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: KS THARUNESHWAR

Question No.	Problem	Constraint Analysis	Solution Development	Application of	Justification	Total Marks (Each – 50)
--------------	---------	---------------------	----------------------	----------------	---------------	-------------------------

	Understanding (25%)	(25%)	t	Concepts	(10%)	
--	--------------------------------	--------------	----------	-----------------	--------------	--

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Marks Evaluation:

			(25%)	(15%)		
Q1						
Q2						
Overall Total (100)						
Faculty In-charge		Course Coordinator		Head of Department		
Signature: _____		Signature: _____		Signature: _____		
Date: _____		Date: _____		Date: _____		